

Coding Pipeline

Main Workflow

```
initial_classifications = classify_labels(dataset)
conflict_cases = detect_conflicts(initial_classifications)
resolved_conflicts_with_new_rules = resolve_conflicts(conflict_cases)
updated_classifications = reclassify_with_new_rules(dataset,
resolved_conflicts_with_new_rules)
```

Initial Classification

```
Function classify_labels(dataset):
    For each label:
        For each coder (simulating multiple human coders):
            Extract relevant excerpts from data
            Apply LLM to classify data based on label descriptions
            Record classifications (0=absent,1=present,3=undecidable)
            Store results
```

Conflict Resolution

```
Function resolve_conflicts(conflict_cases):
    For each label:
        For each conflict case:
            Conduct multiple rounds of discussion using LLM
            Check if consensus is reached
            If consensus reached:
                Generate new labeling rule based on discussion
            Record final discussion and outcome
```

Re-Classification

```
Function reclassify_with_new_rules(dataset,
resolved_conflicts_with_new_rules):
    For each label:
        Apply LLM for classification using:
            - Original label descriptions
            - New labeling rules from discussions
        Record updated classifications
```